

SYSTEM AND INFORMATION INTEGRITY	SI-2	SI-2	FLAW REMEDIATION	The organization:
SYSTEM AND INFORMATION INTEGRITY	SI-2	SI-2a	FLAW REMEDIATION	Identifies, reports, and corrects information system flaws;
SYSTEM AND INFORMATION INTEGRITY	SI-2	SI-2b	FLAW REMEDIATION	Tests software and firmware updates related to flaw remediation for effectiveness and potential side effects before installation;
SYSTEM AND INFORMATION INTEGRITY	SI-2	SI-2c	FLAW REMEDIATION	Installs security-relevant software and firmware updates within [Assignment: organization-defined time period] of the release of the updates; and

SYSTEM AND INFORMATION INTEGRITY	SI-2	SI-2d	FLAW REMEDIATION	Incorporates flaw remediation into the organizational configuration management process.
SYSTEM AND INFORMATION INTEGRITY	SI-2 (1)	SI-2 (1)	CENTRAL MANAGEMENT	The organization centrally manages the flaw remediation process.
SYSTEM AND INFORMATION INTEGRITY	SI-2 (2)	SI-2 (2)	AUTOMATED FLAW REMEDIATION STATUS	The organization employs automated mechanisms [Assignment: organization-defined frequency] to determine the state of information system components with regard to flaw remediation.
SYSTEM AND INFORMATION INTEGRITY	SI-2 (3)	SI-2 (3)	TIME TO REMEDIATE FLAWS / BENCHMARKS FOR CORRECTIVE ACTIONS	The organization:
SYSTEM AND INFORMATION INTEGRITY	SI-2 (3)	SI-2 (3)(a)	TIME TO REMEDIATE FLAWS / BENCHMARKS FOR CORRECTIVE ACTIONS	Measures the time between flaw identification and flaw remediation; and
SYSTEM AND INFORMATION INTEGRITY	SI-2 (3)	SI-2 (3)(b)	TIME TO REMEDIATE FLAWS / BENCHMARKS FOR CORRECTIVE ACTIONS	Establishes [Assignment: organization-defined benchmarks] for taking corrective actions.
SYSTEM AND INFORMATION INTEGRITY	SI-2 (4)	SI-2 (4)	AUTOMATED PATCH MANAGEMENT TOOLS	[Withdrawn: Incorporated into SI-2].

SYSTEM AND INFORMATION INTEGRITY	SI-2 (5)	SI-2 (5)	AUTOMATIC SOFTWARE / FIRMWARE UPDATES	The organization installs [Assignment: organization-defined security-relevant software and firmware updates] automatically to [Assignment: organization-defined information system components].
SYSTEM AND INFORMATION INTEGRITY	SI-2 (6)	SI-2 (6)	REMOVAL OF PREVIOUS VERSIONS OF SOFTWARE / FIRMWARE	The organization removes [Assignment: organization-defined software and firmware components] after updated versions have been installed.
SYSTEM AND INFORMATION INTEGRITY	SI-3	SI-3	MALICIOUS CODE PROTECTION	The organization:

SYSTEM AND INFORMATION INTEGRITY	SI-3	SI-3a	TIME TO REMEDIATE FLAWS / BENCHMARKS FOR CORRECTIVE ACTIONS	Employs malicious code protection mechanisms at information system entry and exit points to detect and eradicate malicious code;
SYSTEM AND INFORMATION INTEGRITY	SI-3	SI-3b	TIME TO REMEDIATE FLAWS / BENCHMARKS FOR CORRECTIVE ACTIONS	Updates malicious code protection mechanisms whenever new releases are available in accordance with organizational configuration management policy and procedures;

SYSTEM AND INFORMATION INTEGRITY	SI-3	SI-3c	TIME TO REMEDIATE FLAWS / BENCHMARKS FOR CORRECTIVE ACTIONS	Configures malicious code protection mechanisms to:
SYSTEM AND INFORMATION INTEGRITY	SI-3	SI-3c.1	TIME TO REMEDIATE FLAWS / BENCHMARKS FOR CORRECTIVE ACTIONS	Perform periodic scans of the information system [Assignment: organization-defined frequency] and real-time scans of files from external sources at [Selection (one or more); endpoint; network entry/exit points] as the files are downloaded, opened, or executed in accordance with organizational security policy; and

SYSTEM AND INFORMATION INTEGRITY	SI-3	SI-3c.2	TIME TO REMEDIATE FLAWS / BENCHMARKS FOR CORRECTIVE ACTIONS	[Selection (one or more): block malicious code; quarantine malicious code; send alert to administrator; [Assignment: organization-defined action]] in response to malicious code detection; and
SYSTEM AND INFORMATION INTEGRITY	SI-3	SI-3d	TIME TO REMEDIATE FLAWS / BENCHMARKS FOR CORRECTIVE ACTIONS	Addresses the receipt of false positives during malicious code detection and eradication and the resulting potential impact on the availability of the information system.

SYSTEM AND INFORMATION INTEGRITY	SI-3 (1)	SI-3 (1)	CENTRAL MANAGEMENT	The organization centrally manages malicious code protection mechanisms.
SYSTEM AND INFORMATION INTEGRITY	SI-3 (2)	SI-3 (2)	AUTOMATIC UPDATES	The information system automatically updates malicious code protection mechanisms.
SYSTEM AND INFORMATION INTEGRITY	SI-3 (3)	SI-3 (3)	NON-PRIVILEGED USERS	[Withdrawn: Incorporated into AC-6 (10)].
SYSTEM AND INFORMATION INTEGRITY	SI-3 (4)	SI-3 (4)	UPDATES ONLY BY PRIVILEGED USERS	The information system updates malicious code protection mechanisms only when directed by a privileged user.
SYSTEM AND INFORMATION INTEGRITY	SI-3 (5)	SI-3 (5)	PORTABLE STORAGE DEVICES	[Withdrawn: Incorporated into MP-7].
SYSTEM AND INFORMATION INTEGRITY	SI-3 (6)	SI-3 (6)	TESTING / VERIFICATION	The organization:
SYSTEM AND INFORMATION INTEGRITY	SI-3 (6)	SI-3 (6)(a)	TESTING / VERIFICATION	Tests malicious code protection mechanisms [Assignment: organization-defined frequency] by introducing a known benign, non-spreading test case into the information system; and
SYSTEM AND INFORMATION INTEGRITY	SI-3 (6)	SI-3 (6)(b)	TESTING / VERIFICATION	Verifies that both detection of the test case and associated incident reporting occur.
SYSTEM AND INFORMATION INTEGRITY	SI-3 (7)	SI-3 (7)	NONSIGNATURE-BASED DETECTION	The information system implements nonsignature-based malicious code detection mechanisms.

SYSTEM AND INFORMATION INTEGRITY	SI-3 (8)	SI-3 (8)	DETECT UNAUTHORIZED COMMANDS	The information system detects [Assignment: organization-defined unauthorized operating system commands] through the kernel application programming interface at [Assignment: organization-defined information system hardware components] and [Selection (one or more): issues a warning; audits the command execution; prevents the execution of the command].
SYSTEM AND INFORMATION INTEGRITY	SI-3 (9)	SI-3 (9)	AUTHENTICATE REMOTE COMMANDS	The information system implements [Assignment: organization-defined security safeguards] to authenticate [Assignment: organization-defined remote commands].
SYSTEM AND INFORMATION INTEGRITY	SI-3 (10)	SI-3 (10)	MALICIOUS CODE ANALYSIS	The organization:
SYSTEM AND INFORMATION INTEGRITY	SI-3 (10)	SI-3 (10)(a)	MALICIOUS CODE ANALYSIS	Employs [Assignment: organization-defined tools and techniques] to analyze the characteristics and behavior of malicious code; and
SYSTEM AND INFORMATION INTEGRITY	SI-3 (10)	SI-3 (10)(b)	MALICIOUS CODE ANALYSIS	Incorporates the results from malicious code analysis into organizational incident response and flaw remediation processes.

SYSTEM AND INFORMATION INTEGRITY	SI-4	SI-4	INFORMATION SYSTEM MONITORING	The organization:
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SYSTEM AND INFORMATION INTEGRITY	SI-4	SI-4a	INFORMATION SYSTEM MONITORING	Monitors the information system to detect:
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SYSTEM AND INFORMATION INTEGRITY	SI-4	SI-4a.1	INFORMATION SYSTEM MONITORING	Attacks and indicators of potential attacks in accordance with [Assignment: organization- defined monitoring objectives]; and
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SYSTEM AND INFORMATION INTEGRITY	SI-4	SI-4a.2	INFORMATION SYSTEM MONITORING	Unauthorized local, network, and remote connections;
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SYSTEM AND INFORMATION INTEGRITY	SI-4	SI-4b	INFORMATION SYSTEM MONITORING	Identifies unauthorized use of the information system through [Assignment: organization- defined techniques and methods];
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SYSTEM AND INFORMATION INTEGRITY	SI-4	SI-4c	INFORMATION SYSTEM MONITORING	Deploys monitoring devices:
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SYSTEM AND INFORMATION INTEGRITY	SI-4	SI-4c.1	INFORMATION SYSTEM MONITORING	Strategically within the information system to collect organization-determined essential information; and
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SYSTEM AND INFORMATION INTEGRITY	SI-4	SI-4c.2	INFORMATION SYSTEM MONITORING	At ad hoc locations within the system to track specific types of transactions of interest to the organization;
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SYSTEM AND INFORMATION INTEGRITY	SI-4	SI-4d	INFORMATION SYSTEM MONITORING	Protects information obtained from intrusion- monitoring tools from unauthorized access, modification, and deletion;
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SYSTEM AND INFORMATION INTEGRITY	SI-4	SI-4e	INFORMATION SYSTEM MONITORING	Heightens the level of information system monitoring activity whenever there is an indication of increased risk to organizational operations and assets, individuals, other organizations, or the Nation based on law enforcement information, intelligence information, or other credible sources of information;
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SYSTEM AND INFORMATION INTEGRITY	SI-4	SI-4f	INFORMATION SYSTEM MONITORING	Obtains legal opinion with regard to information system monitoring activities in accordance with applicable federal laws, Executive Orders, directives, policies, or regulations; and
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SYSTEM AND INFORMATION INTEGRITY	SI-4	SI-4g	INFORMATION SYSTEM MONITORING	Provides [Assignment: organization-defined information system monitoring information] to [Assignment: organization-defined personnel or roles] [Selection (one or more): as needed; [Assignment: organization-defined frequency]].
SYSTEM AND INFORMATION INTEGRITY	SI-4 (1)	SI-4 (1)	SYSTEM-WIDE INTRUSION DETECTION SYSTEM	The organization connects and configures individual intrusion detection tools into an information system-wide intrusion detection system.

SYSTEM AND INFORMATION INTEGRITY	SI-4 (2)	SI-4 (2)	AUTOMATED TOOLS FOR REAL-TIME ANALYSIS	The organization employs automated tools to support near real-time analysis of events.
SYSTEM AND INFORMATION INTEGRITY	SI-4 (3)	SI-4 (3)	AUTOMATED TOOL INTEGRATION	The organization employs automated tools to integrate intrusion detection tools into access control and flow control mechanisms for rapid response to attacks by enabling reconfiguration of these mechanisms in support of attack isolation and elimination.
SYSTEM AND INFORMATION INTEGRITY	SI-4 (4)	SI-4 (4)	INBOUND AND OUTBOUND COMMUNICATIONS TRAFFIC	The information system monitors inbound and outbound communications traffic [Assignment: organization-defined frequency] for unusual or unauthorized activities or conditions.
SYSTEM AND INFORMATION INTEGRITY	SI-4 (5)	SI-4 (5)	SYSTEM-GENERATED ALERTS	The information system alerts [Assignment: organization-defined personnel or roles] when the following indications of compromise or potential compromise occur: [Assignment: organization-defined compromise indicators].
SYSTEM AND INFORMATION INTEGRITY	SI-4 (6)	SI-4 (6)	RESTRICT NON-PRIVILEGED USERS	[Withdrawn: Incorporated into AC-6 (10)].
SYSTEM AND INFORMATION INTEGRITY	SI-4 (7)	SI-4 (7)	AUTOMATED RESPONSE TO SUSPICIOUS EVENTS	The information system notifies [Assignment: organization-defined incident response personnel (identified by name and/or by role)] of detected suspicious events and takes [Assignment: organization-defined least-disruptive actions to terminate suspicious events].
SYSTEM AND INFORMATION INTEGRITY	SI-4 (8)	SI-4 (8)	PROTECTION OF MONITORING INFORMATION	[Withdrawn: Incorporated into SI-4].
SYSTEM AND INFORMATION INTEGRITY	SI-4 (9)	SI-4 (9)	TESTING OF MONITORING TOOLS	The organization tests intrusion-monitoring tools [Assignment: organization-defined frequency].

SYSTEM AND INFORMATION INTEGRITY	SI-4 (10)	SI-4 (10)	VISIBILITY OF ENCRYPTED COMMUNICATIONS	The organization makes provisions so that [Assignment: organization-defined encrypted communications traffic] is visible to [Assignment: organization-defined information system monitoring tools].
SYSTEM AND INFORMATION INTEGRITY	SI-4 (11)	SI-4 (11)	ANALYZE COMMUNICATIONS TRAFFIC ANOMALIES	The organization analyzes outbound communications traffic at the external boundary of the information system and selected [Assignment: organization-defined interior points within the system (e.g., subnetworks, subsystems)] to discover anomalies.
SYSTEM AND INFORMATION INTEGRITY	SI-4 (12)	SI-4 (12)	AUTOMATED ALERTS	The organization employs automated mechanisms to alert security personnel of the following inappropriate or unusual activities with security implications: [Assignment: organization-defined activities that trigger alerts].
SYSTEM AND INFORMATION INTEGRITY	SI-4 (13)	SI-4 (13)	ANALYZE TRAFFIC / EVENT PATTERNS	The organization:
SYSTEM AND INFORMATION INTEGRITY	SI-4 (13)	SI-4 (13)(a)	ANALYZE TRAFFIC / EVENT PATTERNS	Analyzes communications traffic/event patterns for the information system;
SYSTEM AND INFORMATION INTEGRITY	SI-4 (13)	SI-4 (13)(b)	ANALYZE TRAFFIC / EVENT PATTERNS	Develops profiles representing common traffic patterns and/or events; and
SYSTEM AND INFORMATION INTEGRITY	SI-4 (13)	SI-4 (13)(c)	ANALYZE TRAFFIC / EVENT PATTERNS	Uses the traffic/event profiles in tuning system-monitoring devices to reduce the number of false positives and the number of false negatives.

SYSTEM AND INFORMATION INTEGRITY	SI-4 (14)	SI-4 (14)	WIRELESS INTRUSION DETECTION	The organization employs a wireless intrusion detection system to identify rogue wireless devices and to detect attack attempts and potential compromises/breaches to the information system.
SYSTEM AND INFORMATION INTEGRITY	SI-4 (15)	SI-4 (15)	WIRELESS TO WIRELINE COMMUNICATIONS	The organization employs an intrusion detection system to monitor wireless communications traffic as the traffic passes from wireless to wireline networks.
SYSTEM AND INFORMATION INTEGRITY	SI-4 (16)	SI-4 (16)	CORRELATE MONITORING INFORMATION	The organization correlates information from monitoring tools employed throughout the information system.
SYSTEM AND INFORMATION INTEGRITY	SI-4 (17)	SI-4 (17)	INTEGRATED SITUATIONAL AWARENESS	The organization correlates information from monitoring physical, cyber, and supply chain activities to achieve integrated, organization-wide situational awareness.
SYSTEM AND INFORMATION INTEGRITY	SI-4 (18)	SI-4 (18)	ANALYZE TRAFFIC / COVERT EXFILTRATION	The organization analyzes outbound communications traffic at the external boundary of the information system (i.e., system perimeter) and at [Assignment: organization-defined interior points within the system (e.g., subsystems, subnetworks)] to detect covert exfiltration of information.
SYSTEM AND INFORMATION INTEGRITY	SI-4 (19)	SI-4 (19)	INDIVIDUALS POSING GREATER RISK	The organization implements [Assignment: organization-defined additional monitoring] of individuals who have been identified by [Assignment: organization-defined sources] as posing an increased level of risk.

SYSTEM AND INFORMATION INTEGRITY	SI-4 (20)	SI-4 (20)	PRIVILEGED USERS	The organization implements [Assignment: organization-defined additional monitoring] of privileged users.
SYSTEM AND INFORMATION INTEGRITY	SI-4 (21)	SI-4 (21)	PROBATIONARY PERIODS	The organization implements [Assignment: organization-defined additional monitoring] of individuals during [Assignment: organization-defined probationary period].
SYSTEM AND INFORMATION INTEGRITY	SI-4 (22)	SI-4 (22)	UNAUTHORIZED NETWORK SERVICES	The information system detects network services that have not been authorized or approved by [Assignment: organization-defined authorization or approval processes] and [Selection (one or more): audits; alerts [Assignment: organization-defined personnel or roles]].
SYSTEM AND INFORMATION INTEGRITY	SI-4 (23)	SI-4 (23)	HOST-BASED DEVICES	The organization implements [Assignment: organization-defined host-based monitoring mechanisms] at [Assignment: organization-defined information system components].
SYSTEM AND INFORMATION INTEGRITY	SI-4 (24)	SI-4 (24)	INDICATORS OF COMPROMISE	The information system discovers, collects, distributes, and uses indicators of compromise.
SYSTEM AND INFORMATION INTEGRITY	SI-5	SI-5	SECURITY ALERTS, ADVISORIES, AND DIRECTIVES	The organization:

SYSTEM AND INFORMATION INTEGRITY	SI-5	SI-5a	SECURITY ALERTS, ADVISORIES, AND DIRECTIVES	Receives information system security alerts, advisories, and directives from [Assignment: organization-defined external organizations] on an ongoing basis;
SYSTEM AND INFORMATION INTEGRITY	SI-5	SI-5b	SECURITY ALERTS, ADVISORIES, AND DIRECTIVES	Generates internal security alerts, advisories, and directives as deemed necessary;
SYSTEM AND INFORMATION INTEGRITY	SI-5	SI-5c	SECURITY ALERTS, ADVISORIES, AND DIRECTIVES	Disseminates security alerts, advisories, and directives to: [Selection (one or more): [Assignment: organization-defined personnel or roles]; [Assignment: organization-defined elements within the organization]; [Assignment: organization-defined external organizations]]; and
SYSTEM AND INFORMATION INTEGRITY	SI-5	SI-5d	SECURITY ALERTS, ADVISORIES, AND DIRECTIVES	Implements security directives in accordance with established time frames, or notifies the issuing organization of the degree of noncompliance.

SYSTEM AND INFORMATION INTEGRITY	SI-5 (1)	SI-5 (1)	AUTOMATED ALERTS AND ADVISORIES	The organization employs automated mechanisms to make security alert and advisory information available throughout the organization.
SYSTEM AND INFORMATION INTEGRITY	SI-6	SI-6	SECURITY FUNCTION VERIFICATION	The information system:
SYSTEM AND INFORMATION INTEGRITY	SI-6	SI-6a	SECURITY FUNCTION VERIFICATION	Verifies the correct operation of [Assignment: organization-defined security functions];
SYSTEM AND INFORMATION INTEGRITY	SI-6	SI-6b	SECURITY FUNCTION VERIFICATION	Performs this verification [Selection (one or more): [Assignment: organization-defined system transitional states]; upon command by user with appropriate privilege; [Assignment: organization-defined frequency]];
SYSTEM AND INFORMATION INTEGRITY	SI-6	SI-6c	SECURITY FUNCTION VERIFICATION	Notifies [Assignment: organization-defined personnel or roles] of failed security verification tests; and
SYSTEM AND INFORMATION INTEGRITY	SI-6	SI-6d	SECURITY FUNCTION VERIFICATION	[Selection (one or more): shuts the information system down; restarts the information system; [Assignment: organization-defined alternative action(s)]] when anomalies are discovered.
SYSTEM AND INFORMATION INTEGRITY	SI-6 (1)	SI-6 (1)	NOTIFICATION OF FAILED SECURITY TESTS	[Withdrawn: Incorporated into SI-6].
SYSTEM AND INFORMATION INTEGRITY	SI-6 (2)	SI-6 (2)	AUTOMATION SUPPORT FOR DISTRIBUTED TESTING	The information system implements automated mechanisms to support the management of distributed security testing.

SYSTEM AND INFORMATION INTEGRITY	SI-6 (3)	SI-6 (3)	REPORT VERIFICATION RESULTS	The organization reports the results of security function verification to [Assignment: organization-defined personnel or roles].
SYSTEM AND INFORMATION INTEGRITY	SI-7	SI-7	SOFTWARE, FIRMWARE, AND INFORMATION INTEGRITY	The organization employs integrity verification tools to detect unauthorized changes to [Assignment: organization-defined software, firmware, and information].
SYSTEM AND INFORMATION INTEGRITY	SI-7 (1)	SI-7 (1)	INTEGRITY CHECKS	The information system performs an integrity check of [Assignment: organization-defined software, firmware, and information] [Selection (one or more): at startup; at [Assignment: organization-defined transitional states or security-relevant events]; [Assignment: organization-defined frequency]].
SYSTEM AND INFORMATION INTEGRITY	SI-7 (2)	SI-7 (2)	AUTOMATED NOTIFICATIONS OF INTEGRITY VIOLATIONS	The organization employs automated tools that provide notification to [Assignment: organization-defined personnel or roles] upon discovering discrepancies during integrity verification.
SYSTEM AND INFORMATION INTEGRITY	SI-7 (3)	SI-7 (3)	CENTRALLY-MANAGED INTEGRITY TOOLS	The organization employs centrally managed integrity verification tools.
SYSTEM AND INFORMATION INTEGRITY	SI-7 (4)	SI-7 (4)	TAMPER-EVIDENT PACKAGING	[Withdrawn: Incorporated into SA-12].
SYSTEM AND INFORMATION INTEGRITY	SI-7 (5)	SI-7 (5)	AUTOMATED RESPONSE TO INTEGRITY VIOLATIONS	The information system automatically [Selection (one or more): shuts the information system down; restarts the information system; implements [Assignment: organization-defined security safeguards]] when integrity violations are discovered.

SYSTEM AND INFORMATION INTEGRITY	SI-7 (6)	SI-7 (6)	CRYPTOGRAPHIC PROTECTION	The information system implements cryptographic mechanisms to detect unauthorized changes to software, firmware, and information.
SYSTEM AND INFORMATION INTEGRITY	SI-7 (7)	SI-7 (7)	INTEGRATION OF DETECTION AND RESPONSE	The organization incorporates the detection of unauthorized [Assignment: organization-defined security-relevant changes to the information system] into the organizational incident response capability.
SYSTEM AND INFORMATION INTEGRITY	SI-7 (8)	SI-7 (8)	AUDITING CAPABILITY FOR SIGNIFICANT EVENTS	The information system, upon detection of a potential integrity violation, provides the capability to audit the event and initiates the following actions: [Selection (one or more): generates an audit record; alerts current user; alerts [Assignment: organization-defined personnel or roles]; [Assignment: organization-defined other actions]].
SYSTEM AND INFORMATION INTEGRITY	SI-7 (9)	SI-7 (9)	VERIFY BOOT PROCESS	The information system verifies the integrity of the boot process of [Assignment: organization-defined devices].
SYSTEM AND INFORMATION INTEGRITY	SI-7 (10)	SI-7 (10)	PROTECTION OF BOOT FIRMWARE	The information system implements [Assignment: organization-defined security safeguards] to protect the integrity of boot firmware in [Assignment: organization-defined devices].
SYSTEM AND INFORMATION INTEGRITY	SI-7 (11)	SI-7 (11)	CONFINED ENVIRONMENTS WITH LIMITED PRIVILEGES	The organization requires that [Assignment: organization-defined user-installed software] execute in a confined physical or virtual machine environment with limited privileges.
SYSTEM AND INFORMATION INTEGRITY	SI-7 (12)	SI-7 (12)	INTEGRITY VERIFICATION	The organization requires that the integrity of [Assignment: organization-defined user-installed software] be verified prior to execution.

SYSTEM AND INFORMATION INTEGRITY	SI-7 (13)	SI-7 (13)	CODE EXECUTION IN PROTECTED ENVIRONMENTS	The organization allows execution of binary or machine-executable code obtained from sources with limited or no warranty and without the provision of source code only in confined physical or virtual machine environments and with the explicit approval of [Assignment: organization-defined personnel or roles].
SYSTEM AND INFORMATION INTEGRITY	SI-7 (14)	SI-7 (14)	BINARY OR MACHINE EXECUTABLE CODE	The organization:
SYSTEM AND INFORMATION INTEGRITY	SI-7 (14)	SI-7 (14)(a)	BINARY OR MACHINE EXECUTABLE CODE	Prohibits the use of binary or machine-executable code from sources with limited or no warranty and without the provision of source code; and
SYSTEM AND INFORMATION INTEGRITY	SI-7 (14)	SI-7 (14)(b)	BINARY OR MACHINE EXECUTABLE CODE	Provides exceptions to the source code requirement only for compelling mission/operational requirements and with the approval of the authorizing official.
SYSTEM AND INFORMATION INTEGRITY	SI-7 (15)	SI-7 (15)	CODE AUTHENTICATION	The information system implements cryptographic mechanisms to authenticate [Assignment: organization-defined software or firmware components] prior to installation.
SYSTEM AND INFORMATION INTEGRITY	SI-7 (16)	SI-7 (16)	TIME LIMIT ON PROCESS EXECUTION W/O SUPERVISION	The organization does not allow processes to execute without supervision for more than [Assignment: organization-defined time period].

SYSTEM AND INFORMATION INTEGRITY	SI-8	SI-8	SPAM PROTECTION	The organization:
SYSTEM AND INFORMATION INTEGRITY	SI-8	SI-8a	SPAM PROTECTION	Employs spam protection mechanisms at information system entry and exit points to detect and take action on unsolicited messages; and

SYSTEM AND INFORMATION INTEGRITY	SI-8	SI-8b	SPAM PROTECTION	Updates spam protection mechanisms when new releases are available in accordance with organizational configuration management policy and procedures.
SYSTEM AND INFORMATION INTEGRITY	SI-8 (1)	SI-8 (1)	CENTRAL MANAGEMENT	The organization centrally manages spam protection mechanisms.
SYSTEM AND INFORMATION INTEGRITY	SI-8 (2)	SI-8 (2)	AUTOMATIC UPDATES	The information system automatically updates spam protection mechanisms.
SYSTEM AND INFORMATION INTEGRITY	SI-8 (3)	SI-8 (3)	CONTINUOUS LEARNING CAPABILITY	The information system implements spam protection mechanisms with a learning capability to more effectively identify legitimate communications traffic.
SYSTEM AND INFORMATION INTEGRITY	SI-9	SI-9	INFORMATION INPUT RESTRICTIONS	[Withdrawn: Incorporated into AC-2, AC-3, AC-5, AC-6].
SYSTEM AND INFORMATION INTEGRITY	SI-10	SI-10	INFORMATION INPUT VALIDATION	The information system checks the validity of [Assignment: organization-defined information inputs].

SYSTEM AND INFORMATION INTEGRITY	SI-10 (1)	SI-10 (1)	MANUAL OVERRIDE CAPABILITY	The information system:
SYSTEM AND INFORMATION INTEGRITY	SI-10 (1)	SI-10 (1)(a)	MANUAL OVERRIDE CAPABILITY	Provides a manual override capability for input validation of [Assignment: organization-defined inputs];
SYSTEM AND INFORMATION INTEGRITY	SI-10 (1)	SI-10 (1)(b)	MANUAL OVERRIDE CAPABILITY	Restricts the use of the manual override capability to only [Assignment: organization-defined authorized individuals]; and
SYSTEM AND INFORMATION INTEGRITY	SI-10 (1)	SI-10 (1)(c)	MANUAL OVERRIDE CAPABILITY	Audits the use of the manual override capability.
SYSTEM AND INFORMATION INTEGRITY	SI-10 (2)	SI-10 (2)	REVIEW / RESOLUTION OF ERRORS	The organization ensures that input validation errors are reviewed and resolved within [Assignment: organization-defined time period].
SYSTEM AND INFORMATION INTEGRITY	SI-10 (3)	SI-10 (3)	PREDICTABLE BEHAVIOR	The information system behaves in a predictable and documented manner that reflects organizational and system objectives when invalid inputs are received.
SYSTEM AND INFORMATION INTEGRITY	SI-10 (4)	SI-10 (4)	REVIEW / TIMING INTERACTIONS	The organization accounts for timing interactions among information system components in determining appropriate responses for invalid inputs.
SYSTEM AND INFORMATION INTEGRITY	SI-10 (5)	SI-10 (5)	RESTRICT INPUTS TO TRUSTED SOURCES AND APPROVED FORMATS	The organization restricts the use of information inputs to [Assignment: organization-defined trusted sources] and/or [Assignment: organization-defined formats].
SYSTEM AND INFORMATION INTEGRITY	SI-11	SI-11	ERROR HANDLING	The information system:

SYSTEM AND INFORMATION INTEGRITY	SI-11	SI-11a	ERROR HANDLING	Generates error messages that provide information necessary for corrective actions without revealing information that could be exploited by adversaries; and
SYSTEM AND INFORMATION INTEGRITY	SI-11	SI-11b	ERROR HANDLING	Reveals error messages only to [Assignment: organization-defined personnel or roles].
SYSTEM AND INFORMATION INTEGRITY	SI-12	SI-12	INFORMATION HANDLING AND RETENTION	The organization handles and retains information within the information system and information output from the system in accordance with applicable federal laws, Executive Orders, directives, policies, regulations, standards, and operational requirements.
SYSTEM AND INFORMATION INTEGRITY	SI-13	SI-13	PREDICTABLE FAILURE PREVENTION	The organization:
SYSTEM AND INFORMATION INTEGRITY	SI-13	SI-13a	PREDICTABLE FAILURE PREVENTION	Determines mean time to failure (MTTF) for [Assignment: organization-defined information system components] in specific environments of operation; and
SYSTEM AND INFORMATION INTEGRITY	SI-13	SI-13b	PREDICTABLE FAILURE PREVENTION	Provides substitute information system components and a means to exchange active and standby components at [Assignment: organization-defined MTTF substitution criteria].

SYSTEM AND INFORMATION INTEGRITY	SI-13 (1)	SI-13 (1)	TRANSFERRING COMPONENT RESPONSIBILITIES	The organization takes information system components out of service by transferring component responsibilities to substitute components no later than [Assignment: organization-defined fraction or percentage] of mean time to failure.
SYSTEM AND INFORMATION INTEGRITY	SI-13 (2)	SI-13 (2)	TIME LIMIT ON PROCESS EXECUTION WITHOUT SUPERVISION	[Withdrawn: Incorporated into SI-7 (16)].
SYSTEM AND INFORMATION INTEGRITY	SI-13 (3)	SI-13 (3)	MANUAL TRANSFER BETWEEN COMPONENTS	The organization manually initiates transfers between active and standby information system components [Assignment: organization-defined frequency] if the mean time to failure exceeds [Assignment: organization-defined time period].
SYSTEM AND INFORMATION INTEGRITY	SI-13 (4)	SI-13 (4)	STANDBY COMPONENT INSTALLATION / NOTIFICATION	The organization, if information system component failures are detected:
SYSTEM AND INFORMATION INTEGRITY	SI-13 (4)	SI-13 (4)(a)	STANDBY COMPONENT INSTALLATION / NOTIFICATION	Ensures that the standby components are successfully and transparently installed within [Assignment: organization-defined time period]; and
SYSTEM AND INFORMATION INTEGRITY	SI-13 (4)	SI-13 (4)(b)	STANDBY COMPONENT INSTALLATION / NOTIFICATION	[Selection (one or more): activates [Assignment: organization-defined alarm]; automatically shuts down the information system].
SYSTEM AND INFORMATION INTEGRITY	SI-13 (5)	SI-13 (5)	FAILOVER CAPABILITY	The organization provides [Selection: real-time; near real-time] [Assignment: organization-defined failover capability] for the information system.

SYSTEM AND INFORMATION INTEGRITY	SI-14	SI-14	NON-PERSISTENCE	The organization implements non-persistent [Assignment: organization-defined information system components and services] that are initiated in a known state and terminated [Selection (one or more): upon end of session of use; periodically at [Assignment: organization-defined frequency]].
SYSTEM AND INFORMATION INTEGRITY	SI-14 (1)	SI-14 (1)	REFRESH FROM TRUSTED SOURCES	The organization ensures that software and data employed during information system component and service refreshes are obtained from [Assignment: organization-defined trusted sources].
SYSTEM AND INFORMATION INTEGRITY	SI-15	SI-15	INFORMATION OUTPUT FILTERING	The information system validates information output from [Assignment: organization-defined software programs and/or applications] to ensure that the information is consistent with the expected content.
SYSTEM AND INFORMATION INTEGRITY	SI-16	SI-16	MEMORY PROTECTION	The information system implements [Assignment: organization-defined security safeguards] to protect its memory from unauthorized code execution.
SYSTEM AND INFORMATION INTEGRITY	SI-17	SI-17	FAIL-SAFE PROCEDURES	The information system implements [Assignment: organization-defined fail-safe procedures] when [Assignment: organization-defined failure conditions occur].